

Python lab : 6th July 2026

Lab 1: Student Marks Management

Problem Statement:

Create a dictionary to store the marks of 5 students, where the key is the student's name and the value is their marks.

Perform the following operations:

- Display all student names and marks.
 - Add a new student with marks.
 - Update the marks of an existing student.
 - Delete a student by name.
 - Display the student who scored the highest marks.
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Lab 2: Word Frequency Counter

Problem Statement:

Accept a sentence from the user and create a dictionary that stores the frequency of each word.

Example:

Input:

python is easy and python is powerful

Output:

```
{  
'python': 2,  
'is': 2,  
'easy': 1,  
'and': 1,  
'powerful': 1  
}
```

Additionally:

- Display the most frequently occurring word.
 - Display all words in alphabetical order.
-

Lab 3: Employee Information System

Problem Statement:

Create a dictionary where:

- Employee ID is the key.

- Value is another dictionary containing:
 - Name
 - Department
 - Salary

Perform the following operations:

- Display all employee details.
 - Search for an employee using Employee ID.
 - Increase the salary of all employees by 10%.
 - Display employees belonging to a specific department entered by the user.
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Lab 4: Inventory Management

Problem Statement:

Create a dictionary to maintain the stock of products in a shop.

Example:

```
{  
'Laptop': 15,  
'Mouse': 40,  
'Keyboard': 25,  
'Monitor': 10  
}
```

Implement the following:

- Add a new product.
 - Update the stock of an existing product.
 - Remove a product from inventory.
 - Display products having stock less than 20.
 - Display the total number of items available in the inventory.
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Lab 5: Student Subject Report Card

Problem Statement:

Create a nested dictionary to store marks of students in three subjects.

Example:

```
{  
'Rahul': {'Math': 85, 'Science': 90, 'English': 88},  

```

```
'Priya': {'Math': 78, 'Science': 95, 'English': 82},  
'Ankit': {'Math': 91, 'Science': 89, 'English': 94}  
}
```

Write a program to:

- Calculate the total marks of each student.
- Calculate the average marks of each student.
- Display the topper based on total marks.
- Display the subject-wise highest marks along with the student's name.
- Display students whose average is greater than or equal to 85.